

My Week 1 Research Report

1. Literature Summary

Core Principles of Prompt Engineering and Structured Prompting

In my research, I define prompt engineering as the practice of structuring text inputs so that Generative AI models produce accurate, high-quality, and context-aware responses. In educational setups, I have found that structured prompting ensures the AI behaves like a trained educator rather than a generic text generator.

I have identified several key techniques:

- **Role Prompting:** Assigning a specific persona to the AI (e.g., "*Act as a patient computer science tutor*"). This sets the tone, depth of knowledge, and boundaries for the interaction.
- **Few-Shot Prompting:** Providing explicit input-output examples before asking the final question. This demonstrates the desired format, style, and quality standard to the AI.
- **Chain-of-Thought (CoT) Prompting:** Instructing the AI to display its step-by-step reasoning process before giving the final answer. This reduces logical errors, especially in complex problem-solving.
- **Context Layering:** Building multi-layered prompts that isolate background information, core constraints, target audience level, and output formatting rules into distinct sections.

Current Applications of AI in Digital Learning Environments

I am observing AI moving from a passive search tool to an active participant in digital education. From my perspective, AI systems are currently applied in three main areas:

- **Intelligent Tutoring Systems (ITS):** Providing 24/7 chat-based support where students can ask questions about course content and receive instant explanations.
- **Automated Formative Assessment:** Reviewing student drafts, code submissions, or open-ended answers to provide immediate feedback before final human grading.
- **Operational Automation:** Assisting instructors by drafting lesson plans, generating quiz variations, and sorting discussion forum posts by urgency.

Research Insights on Personalization and Structured Pathways

Through my review of educational research, I want to emphasize that learning is most effective when it matches a student's current cognitive state.

- **Cognitive Load Theory (CLT):** Human working memory is limited. My research shows that unguided or unstructured learning pathways overwhelm students. I believe AI can solve this by breaking complex information into smaller, bite-sized modules based on the student's real-time understanding.

- **Zone of Proximal Development (ZPD):** Content should be neither too easy (causing boredom) nor too hard (causing frustration). Modern AI personalization uses continuous tracking to keep learning material exactly within this optimal challenge zone by adjusting text difficulty and scaffolding support.

AI in Opportunity Creation and Content Generation

I have noted that AI significantly speeds up the time it takes to build and launch educational programs. In the modern career ecosystems I am studying, AI tools analyze real-time job market data to generate matching course modules.

Additionally, I see AI supporting "opportunity creation" by translating abstract learning goals into concrete professional challenges. I have found it can ingest complex company case studies and automatically generate structured student internship briefs, project milestones, and evaluation scorecards, bridging the gap between education and employment.

2. Key Problem Identification

Problem 1: AI-Generated Content Inaccuracy and "Hallucinations"

Description: AI models frequently generate text that sounds highly confident but is factually incorrect, outdated, or logically flawed.

- **My Research Insight & Impact:** My research into education shows that inaccurate materials destroy learner trust and create dangerous misconceptions. I am concerned that if a learner cannot verify whether an AI tutor's explanation is correct, the platform's educational credibility fails. This leads me to conclude that unverified AI content generation is a high risk for core curriculum design.

Problem 2: The Engagement Gap in Non-Interactive Personalization

Description: Many AI personalization systems simply dump customized reading materials on a student's dashboard without creating an interactive loop or emotional connection.

- **My Research Insight & Impact:** Based on my understanding of self-determination theory, online learners need a sense of belonging and competence to stay motivated. I find that static personalization feels mechanical. I've observed that without dynamic, conversational elements or community connections, learners experience isolation, leading to high course dropout rates.

Problem 3: Disconnected Learning Pathways and Skill Fragmentation

Description: AI tools often recommend separate, isolated lessons based on immediate mistakes, failing to connect these pieces into a long-term, career-ready

roadmap.

- **My Research Insight & Impact:** I believe effective skill development requires cumulative learning, where each new concept builds clearly on the last. When I see AI pathways that are too fragmented, I worry that students master isolated micro-skills but lack the big-picture capability required to complete comprehensive, real-world portfolio projects.

3. My Structured Use-Case Proposals

Use Case 1: The Adaptive Socratic Tutor

- **Target Area:** Learning Assistant Feature
- **The Problem I Identified:** I have noticed that students often copy and paste assignment questions into AI tools to get direct answers, bypassing the actual critical thinking and learning process.
- **My Proposed Solution:** I propose a chat-based assistant programmed with strict negative constraints: it is forbidden from providing direct solutions. Instead, I want it to use **Chain-of-Thought** and **Role Prompting** to ask guiding, step-by-step questions based on the student's incorrect input, leading them to discover the answer themselves.
- **Expected Benefit:** Boosts active learning, prevents academic dishonesty, and helps students develop true problem-solving skills.

Use Case 2: The Real-Time Skill Gap Router

- **Target Area:** Structured Learning Pathways
- **The Problem I Identified:** I've seen that students discover they are missing foundational skills only *after* failing a major project milestone, causing frustration and delays.
- **My Proposed Solution:** I suggest an automated diagnostic tool that reviews project drafts. If a specific error is found, I want the AI to instantly pause the project timeline and generate a 30-minute mini-learning pathway to patch that specific skill gap.
- **Expected Benefit:** Reduces student frustration by providing help exactly when it is needed, preventing project failure.

Use Case 3: Automated Experiential Project Brief Builder

- **Target Area:** Content Generation & Opportunity Creation
- **The Problem I Identified:** I have observed that transforming raw corporate problems into structured, educational project briefs for student internships takes human program managers days or weeks.
- **My Proposed Solution:** I propose a structured workflow template where employer partners input raw company goals. My system would use **Context Layering** to output a complete, student-ready package: weekly milestones, clear deliverables, and detailed performance rubrics.

- **Expected Benefit:** Allows educational platforms to scale their internship and masterclass offerings rapidly, opening up more real-world work opportunities for students.

Use Case 4: AI Visual Milestone Badge System

- **Target Area:** Visual Identity & Motivation
- **The Problem I Identified:** I believe that text-based notifications for finishing a course module fail to provide a lasting sense of accomplishment or professional pride.
- **My Proposed Solution:** I've conceptualized an integrated AI system that combines text analysis of a student's project highlights with image-generation parameters. When a student completes a specialized pathway, I want the AI to automatically generate a customized, visually distinct digital achievement badge and micro-credential icon tailored to that specific project theme.
- **Expected Benefit:** Increases student motivation through visual rewards and provides clean, shareable assets for students to display on professional networks.